

Zero-Trust Shielded Graph-Structured Decision Control for Secure Autonomous Recovery in AI-Native 6G Disaster-Monitoring Sensing Networks

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Abstract—Disaster-monitoring sensing networks require autonomous recovery under adversarial conditions, yet no existing framework combines graph-structured decision making with formally bounded zero-trust admissibility for near-real-time actuation under attack. We propose *CASTER-ZT*, coupling a graph neural network encoder, learned recovery policy, and neural dual-hypothesis trust–risk evaluator with a deterministic shield that enforces bounded admissibility before execution. Ten formal properties are proved, including a defense-in-depth bound, a distribution-free conformal coverage guarantee, and a calibration convergence rate. A 2,420-run campaign across eleven methods, four attack conditions, two severity levels, twenty seeds, and three scales (12/36/100-cell) shows 74.2% rogue detection with zero false positives at 12-cell (rising to 98.3% at 100-cell), recovery quality 0.733 vs. 0.14–0.20 for non-shielding baselines (p less than 0.001), and 3.07 ms per-decision latency within the O-RAN 10 ms budget.

Index Terms—Zero-trust actuation, graph neural network, autonomous disaster recovery, 6G, shielded autonomy.

I. INTRODUCTION

AUTONOMOUS recovery is becoming a defining requirement in next-generation sensing and communication infrastructures. In disaster-monitoring sensing networks, large-scale failure, degraded observability, and rapidly changing priorities make purely manual recovery too slow. Future AI-native systems must make near-real-time decisions on routing, reassignment, and restoration under uncertainty, yet the same autonomy that speeds recovery enlarges the attack surface when telemetry is corrupted or identities are abused [1], [2].

Graph learning provides a natural formalism for networked recovery [3]–[5], while the safe-learning literature shows that unconstrained policies are unacceptable in safety-sensitive settings [6]–[8]. Parallel work on calibration, uncertainty, and selective prediction emphasizes that accuracy alone is insufficient when the system must know when *not* to act autonomously [9], [10]. Agentic and intent-driven AI for O-RAN and next-generation control further expands autonomous operation [11], [12], but prioritizes orchestration flexibility over formally bounded security under corrupted evidence.

The central gap is that current methods optimize recovery quality *before* defining a formal admissibility boundary for autonomous action under compromised or uncertain evidence—a

dangerous ordering in disaster settings. This paper addresses that gap by formulating post-disaster recovery as a graph-structured secure decision problem with explicit trust, risk, uncertainty, and admissibility constraints. We propose *CASTER-ZT* (Causal, Autonomous, Shielded, Telemetry- and Event-aware Recovery with Zero Trust), a framework in which a recovery policy proposes candidate actions, a dual-hypothesis trust–risk evaluator estimates action safety under benign and adversarial hypotheses, and a formal zero-trust shield [2] determines whether the action may be allowed, scope-reduced, deferred, escalated, or blocked.

The paper asks four research questions. **RQ1**: Can a zero-trust shielded policy reduce adversarial degradation of autonomous recovery? **RQ2**: What security–utility–overhead tradeoffs arise from trust-, authorization-, and risk-aware shielding? **RQ3**: Does the framework remain effective across attack severities and topology scales? **RQ4**: Does the model satisfy 6G near-real-time deployment constraints?

Contributions. (C1) We formulate secure autonomous recovery as a graph-structured decision problem with zero-trust admissibility constraints. (C2) We introduce *CASTER-ZT*, coupling a trained GNN encoder, learned policy, neural trust–risk evaluator, MC-Dropout uncertainty estimation, and a formally verified deterministic shield. (C3) We prove ten bounded properties including defense-in-depth, conformal coverage, and calibration convergence. (C4) A 2,420-run campaign across three scales, four external baselines, and four ablations demonstrates 74.2–98.3% rogue detection with zero false positives, $\omega_{\text{rec}} = 0.733$ vs. 0.14–0.20 for non-shielding baselines, and 3.07 ms per-decision latency. (C5) Component ablation confirms each shield module’s individual necessity.

The rest of the paper is organized as follows. Section II reviews related work. Section III presents the *CASTER-ZT* framework, algorithms, training pipeline, and theoretical properties. Section IV details the experimental evaluation. Section V discusses findings and limitations. Section VI concludes.

II. RELATED WORK

Graph-structured learning for networks. Graph Neural Networks (GNNs) have transformed network representation, from spectral methods [3] and inductive approaches [4] to attention mechanisms [5] and message-passing frameworks [13], [14].

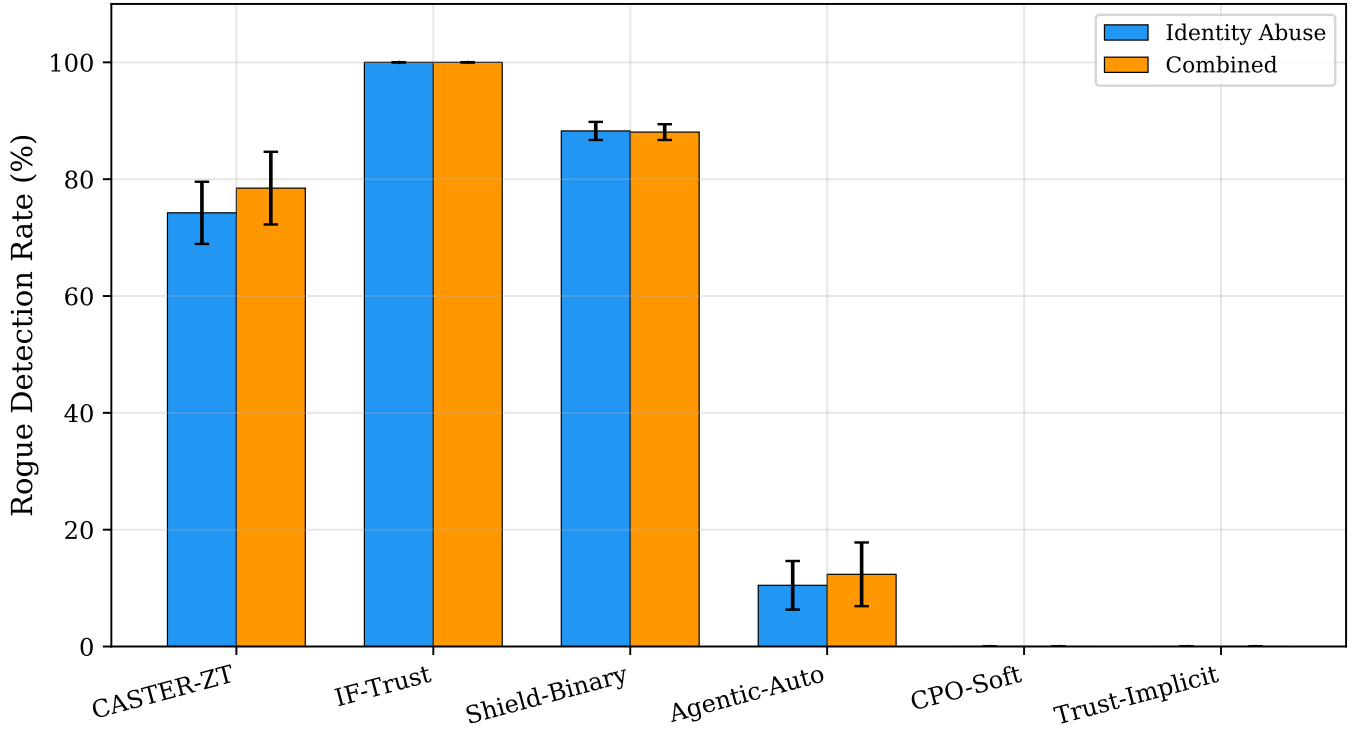


Fig. 1. Rogue detection (RogueDet) and blocking (RogueBlk) under HIGH-severity identity abuse. CASTER-ZT: 74.2% detection, zero false positives; IF-Trust: 100% via aggressive scope-reduction; Shield-Binary: 88.3% with 48.8% false positives.

Temporal GNNs [15], [16] extend these to dynamic graphs. Almasan et al. [17] and Raftopoulos and Fokianakis [18] demonstrate GNN-based routing and control, while Rusek et al. [19] model network QoS via GNNs. However, these approaches optimize performance without security-first actuation gating.

Safe and shielded learning. The safe reinforcement learning (RL) literature addresses constraint satisfaction during training and deployment [6], [20]. Constrained policy optimization [7] bounds cumulative cost; Safe RLHF [21] extends these ideas to language-model alignment. Alshiekh et al. [8] introduce pre-computed binary safety shields from temporal specifications. Roderick et al. [22] propose dynamic model predictive shielding, separating learned perception from deterministic safety. These methods do not incorporate identity verification, trust assessment, or graduated (multi-outcome) shield decisions for security-critical networks.

Uncertainty, calibration, and anomaly detection. MC-Dropout [9] provides scalable epistemic uncertainty; conformal prediction [10] provides distribution-free coverage guarantees. Autoencoders [23], [24] and isolation forests [25], [26] are standard anomaly detectors. Selective prediction [27] and temperature scaling [28] further enable calibrated abstention. These components are individually valuable but have not been composed into a multi-gate zero-trust shield for autonomous network recovery.

Resilient sensing and zero-trust architecture. Sensor-network resilience has been studied via real deployments [29], [30] and cyber-physical security testbeds [31]–[33]. Zero-trust architecture replaces perimeter-based security with continuous

TABLE I
MODEL-SPACE POSITIONING. ✓ = ADDRESSED; ○ = PARTIAL; ✗ = ABSENT.

Family	Graph	ZT	Trust	Shield	Unc.
GNN routing [17], [19]	✓	✗	✗	✗	✗
Safe RL [7], [21]	✗	✗	✗	○	✗
Formal shield [8], [22]	✗	✗	✗	✓	✗
Anomaly trust [25], [26]	✗	○	✓	✗	✗
Agentic AI [11], [43]	○	✗	✗	✗	○
CASTER-ZT	✓	✓	✓	✓	✓

per-action verification [2], [34], [35]. Adversarial attacks on GNNs [36], [37] highlight vulnerabilities in graph-based systems. None of these works combines trust-assessed, graph-structured, formally shielded autonomous recovery.

AI-native 6G and agentic AI. The 6G vision positions AI as a first-class architectural element [38]–[42]. Agentic AI architectures [11], [43], [44] and edge/intent-driven solutions [45]–[47] expand autonomous network control. The EU AI Act [48], NIST AI Risk Management [49], and OWASP Top 10 for LLMs [50] highlight governance requirements. These systems prioritize orchestration flexibility but lack formally bounded security-first actuation under corrupted evidence.

Positioning. Table I positions CASTER-ZT against the literature. The combination of graph-structured state, zero-trust admissibility with graduated multi-outcome decisions, dual-hypothesis trust–risk evaluation, and uncertainty-based abstention is not replicated by any individual model family.

III. CASTER-ZT FRAMEWORK

A. System and threat model

Networked disaster-recovery setting. The system is an autonomous decision controller deployed at the edge/gateway tier of a post-disaster sensing and communication network. The network is modeled as a time-varying directed graph $G_t = (V_t, E_t, X_t)$, where V_t are sensing/communication cells, E_t are connectivity links, and X_t is the real-valued feature matrix. Each cell $v \in V_t$ has an operational state in {operational, degraded, failed, recovering} and five input features (operational state, throughput, latency, loss rate, zone ID). Edges carry two features (link capacity, link utilization). A disaster onset at tick 3 causes $\approx 40\%$ simultaneous cell failure. The controller must restore the network by selecting recovery actions $a_t \in \mathcal{A}$ from six types: POWER-CYCLE, REROUTE-TRAFFIC, REASSIGN-FREQUENCY, CELL-RECONFIG, ISOLATE-CELL, ACTIVATE-BACKUP.

Threat model. An adversary operates under a bounded budget affecting a single zone. Four attack conditions are tested: CLEAN (no attack), TELEMETRY POISONING (corrupted sensor readings), IDENTITY-CREDENTIAL ABUSE (stolen legitimate credentials used to inject rogue recovery actions), and COMBINED. Each is tested at MEDIUM and HIGH severity.

Decision space.

Definition 1 (Bounded decision space). *The shield produces decisions $d_t \in \mathcal{D}$, ordered by conservatism from least to most restrictive:*

$$\mathcal{D} = \{\text{ALLOW}, \text{SCOPE-REDUCE}, \text{DEFER}, \text{ESCALATE}, \text{BLOCK}\}.$$

Optimization objective. The controller maximizes recovery utility subject to the constraint that every action must pass the zero-trust shield:

$$\begin{aligned} \max_{\pi_\theta} \mathbb{E} \left[\sum_t U(a_t^*, s_t) - \lambda_1 U_{\text{sec}}(a_t^*, s_t) \right. \\ \left. - \lambda_2 U_{\text{lat}}(a_t^*, s_t) - \lambda_3 U_{\text{ener}}(a_t^*, s_t) \right] \\ \text{s.t. } a_t^* \in \mathcal{E}(d_t, \hat{a}_t). \end{aligned}$$

Definition 2 (6G Near-Real-Time Compliance). *A system is 6G near-real-time compliant if: (1) per-decision latency $\ell_d \leq 10 \text{ ms}$ (O-RAN Near-RT RIC budget [12], [51]); (2) all inference executes locally without cloud dependency [18]; (3) total model memory $M_{\text{model}} \leq M_{\text{edge}}$ ($\sim 2\text{--}50 \text{ MB}$) [12], [46].*

Definition 3 (AI-Native Pipeline). *A pipeline is AI-native if every stage is a trained ML component. AI-native coverage $\eta_{\text{AI}} = |\{S_k : S_k \text{ is learned}\}|/K$. CASTER-ZT achieves $\eta_{\text{AI}} = 5/6$: five neural stages plus one deliberately deterministic shield.*

B. Architecture

Design principle. *The policy proposes; the zero-trust shield decides whether execution is admissible.* AI-native coverage ($\eta_{\text{AI}} = 5/6$): every stage except the shield is a trained neural

module. 6G compliance: $\approx 20\text{K}$ parameters, $\leq 2 \text{ MB}$, 2-layer GNN satisfying O-RAN latency and memory constraints.

End-to-end dataflow. $\text{telemetry} \rightarrow \underbrace{\text{GNN enc.}}_{\theta} \rightarrow \underbrace{\text{policy}}_{\theta} \rightarrow \underbrace{\text{trust-risk}}_{\phi} \rightarrow \underbrace{\text{MC-Drop}}_{\omega} \rightarrow \underbrace{\text{ZT shield}}_{\text{det.}} \rightarrow \text{bounded exec.}$

The architecture contains five layers, of which the first four are learned:

- 1) **GNN graph encoder** (f_θ , learned). 2-layer GraphSAGE [4] with mean aggregation, $d = 64$, ReLU. Input: 5 node features, 2 edge features. Produces $h_v^{(L)} \in \mathbb{R}^{64}$ via message passing and a graph-level readout $\bar{h}_{G_t}$. Total: $\approx 4.5\text{K}$ params.
- 2) **Recovery policy head** (π_θ , learned). 2-layer MLP mapping $[h_a \| \bar{h}_{G_t}]$ to action logits, softmax over $\mathcal{A}$. Trained via supervised imitation.
- 3) **Dual-hypothesis trust-risk evaluator** (Q_ϕ , learned). Autoencoder anomaly detector for trust (τ_t): reconstruction error $e_t = \|x_t - \text{AE}_\phi(x_t)\|_2$, trust $\tau_t = \sigma(-\beta_\tau(e_t - e_{\text{thresh}}))$. Contrastive safety encoder for divergence (Δ_t). Neural risk scorer for ρ_t .
- 4) **MC-Dropout uncertainty** (U_ω , learned). $T = 20$ stochastic forward passes; $u_t = T^{-1} \sum_i \text{Var}[\pi_\theta^{(i)}(a|s_t)]$.
- 5) **Zero-trust shield** (Γ , deterministic). Formal admissibility gate (Algorithm 1) with 14 scalar parameters set by offline calibration. No learned weights.

C. Shield design and algorithms

1) **Shield algorithm: Complexity.** Total per-decision: $\mathcal{O}(L|E_t|d + |\mathcal{A}|k)$.

2) **Shield calibration:** Algorithm 2 sets shield thresholds via offline calibration on n annotated episodes.

D. Training pipeline

Training data is generated from 2,000 simulated disaster-recovery episodes spanning all four attack conditions, producing $\approx 60,000$ annotated decision steps (70/15/15 train/val/test split). All architectural hyperparameters satisfy the 6G deployment constraints of Definition 2: GNN depth ($L = 2$), latent dimension ($d = 64$), and autoencoder bottleneck ($d_{\text{in}} \rightarrow 32 \rightarrow 16 \rightarrow 8$) are sized to keep total parameters under 50K and memory under 2 MB.

GNN encoder and policy. Trained jointly via supervised imitation with cross-entropy loss, Adam optimizer ($\eta = 10^{-3}$), batch size 64, early stopping (patience 15). Dropout ($p = 0.1$) enables MC-Dropout uncertainty at inference.

Trust autoencoder. Trained on clean telemetry with MSE reconstruction loss ($\eta = 5 \times 10^{-4}$). The 95th-percentile reconstruction error sets e_{thresh} .

Risk scorer and contrastive encoder. Risk scorer: binary cross-entropy on labeled outcomes. Contrastive encoder: margin loss ($m = 1.0$) pulling benign/adversarial hypotheses apart for unsafe actions.

E. Theoretical properties

We state ten propositions; all proofs are provided in the supplemental material.

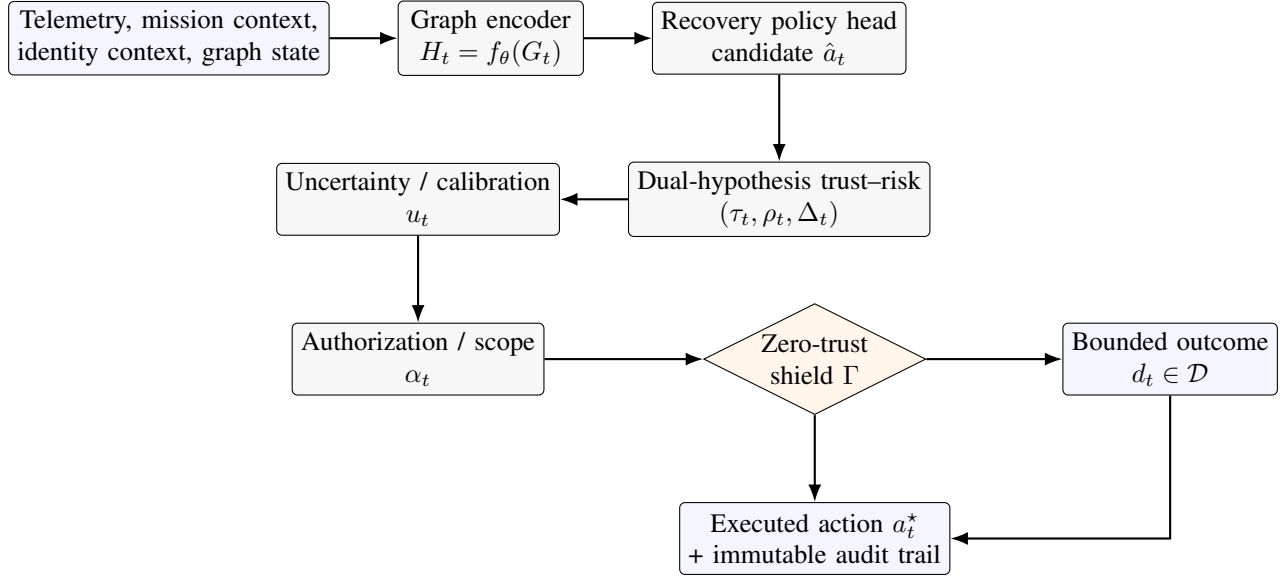


Fig. 2. CASTER-ZT architecture. The policy proposes; the zero-trust shield Γ controls execution via trust, risk, divergence, authorization, and uncertainty signals.

Algorithm 1 CASTER-ZT bounded autonomous recovery decision

Require: state s_t , candidate action set $\mathcal{A}(s_t)$

- 1: Encode graph: $H_t \leftarrow f_\theta(G_t)$
- 2: Score candidates: $\pi_\theta(a | s_t)$ for $a \in \mathcal{A}(s_t)$
- 3: Select top: $\hat{a}_t \leftarrow \arg \max \pi_\theta(a | s_t)$
- 4: Authorization: $\alpha_t \leftarrow A(i_t, \hat{a}_t, s_t)$
- 5: Trust: $\tau_t \leftarrow T_\phi(s_t, \tilde{x}_t^{(0)})$
- 6: Risk: $\rho_t \leftarrow R_\phi(\hat{a}_t, s_t)$
- 7: Divergence: $\Delta_t \leftarrow \|q_\phi(\hat{a}_t, s_t, \tilde{x}_t^{(0)}) - q_\phi(\hat{a}_t, s_t, \tilde{x}_t^{(1)})\|_2$
- 8: Uncertainty: $u_t \leftarrow U_\omega(s_t, \hat{a}_t)$
- 9: Threshold: $\tau_{\min} \leftarrow \tau_0 + \kappa_1 \rho_t + \kappa_2 u_t$
- 10: Conservatism: $g_t \leftarrow w_1(1 - \tau_t) + w_2 \rho_t + w_3 u_t + w_4 \Delta_t$
- 11: **if** $\alpha_t = 0$ **then**
- 12: $d_t \leftarrow \text{BLOCK}$
- 13: **else if** $\Delta_t > \delta_{\max}^{\text{hard}}$ **then**
- 14: $d_t \leftarrow \text{BLOCK}$
- 15: **else if** $\tau_t \geq \tau_{\min}$ **and** $g_t < \gamma_1$ **and** $\Delta_t \leq \delta_{\max}$ **then**
- 16: $d_t \leftarrow \text{ALLOW}$
- 17: **else if** $\gamma_1 \leq g_t < \gamma_2$ **then**
- 18: $d_t \leftarrow \text{SCOPE-REDUCE}$
- 19: **else if** $\gamma_2 \leq g_t < \gamma_3$ **then**
- 20: $d_t \leftarrow \text{DEFER}$
- 21: **else if** $\gamma_3 \leq g_t < \gamma_4$ **or** $u_t > u_{\max}$ **then**
- 22: $d_t \leftarrow \text{ESCALATE}$
- 23: **else**
- 24: $d_t \leftarrow \text{BLOCK}$
- 25: **end if**
- 26: **return** d_t , executed action $a_t^* = \mathcal{E}(d_t, \hat{a}_t)$, audit record

Assumption 1 (Standing assumptions). (1) $w_1, w_2, w_3, w_4 > 0$; (2) $\kappa_1, \kappa_2 \geq 0$; (3) $0 < \gamma_1 < \gamma_2 < \gamma_3 < \gamma_4$; (4) $0 < \delta_{\max} < \delta_{\max}^{\text{hard}}$; (5) $|\mathcal{A}| < \infty$; (6) $\tau_t, \rho_t, u_t \in [0, 1]$, $\Delta_t \geq 0$.

Proposition 1 (Monotone shield conservatism). Under As-

Algorithm 2 Shield threshold calibration

Require: Calibration set $\mathcal{D}_{\text{cal}}$ of n episodes, target $\varepsilon_{\text{target}}$, confidence $1 - \delta$

- 1: Grid: $\Gamma_{\text{grid}} \leftarrow \{(\gamma_1, \gamma_2, \gamma_3, \gamma_4) : 0 < \gamma_1 < \gamma_2 < \gamma_3 < \gamma_4\}$
- 2: **for each** $(\gamma_1, \gamma_2, \gamma_3, \gamma_4) \in \Gamma_{\text{grid}}$ **do**
- 3: Run shield Γ on $\mathcal{D}_{\text{cal}}$
- 4: $\hat{\varepsilon} \leftarrow n^{-1} \sum_{i=1}^n \mathbf{1}[d_i = \text{ALLOW} \wedge y_i = \text{unsafe}]$
- 5: $\varepsilon_{\text{ub}} \leftarrow \hat{\varepsilon} + \sqrt{\ln(1/\delta)/(2n)}$
- 6: **if** $\varepsilon_{\text{ub}} \leq \varepsilon_{\text{target}}$ **then**
- 7: Record feasible configuration
- 8: **end if**
- 9: **end for**
- 10: Select feasible configuration minimizing false-block rate
- 11: **return** calibrated $(\gamma_1^*, \gamma_2^*, \gamma_3^*, \gamma_4^*)$

TABLE II
TRAINING SUMMARY FOR ALL LEARNED COMPONENTS.

Component	Params	Loss	Epochs
GraphSAGE encoder (f_θ)	$\approx 4.5\text{K}$	Cross-entropy	15–20
Policy head (π_θ)	$\approx 8.5\text{K}$	Cross-entropy	15–20
Trust AE (AE_ϕ)	$\approx 1.8\text{K}$	MSE recon.	10–20
Risk scorer (R_ϕ)	$\approx 1.5\text{K}$	Binary CE	25–30
Contrastive enc. (q_ϕ)	$\approx 3.6\text{K}$	Contrastive	40–50
Total learned	$\approx 20\text{K}$	—	—
Shield (deterministic)	14 scalars	Calibration	—

sumption 1 with $\alpha_t = 1$, decreasing τ_t or increasing ρ_t , u_t , or Δ_t cannot move the output to a less conservative decision.

Proposition 2 (Unauthorized-actuation exclusion). If $\alpha_t = 0$, then $d_t \neq \text{ALLOW}$.

Proposition 3 (Divergence-triggered safeguard). If $\Delta_t > \delta_{\max}^{\text{hard}}$, then $d_t = \text{BLOCK}$. If $\Delta_t > \delta_{\max}$, then $d_t \neq \text{ALLOW}$.

Proposition 4 (Bounded decision completeness). Γ is a total

TABLE III
COMPLETE HYPERPARAMETER SPECIFICATION.

Component	Parameter	Value
GNN encoder	Layers L	2
	Latent dim. d	64
Training	Optimizer	Adam
	LR (GNN, policy, risk)	10^{-3}
	LR (autoencoder)	5×10^{-4}
	Batch size	64
Uncertainty	Dropout p	0.1
	MC passes T	20
Shield (Γ)	$\gamma_1, \gamma_2, \gamma_3, \gamma_4$	0.30, 0.50, 0.70, 0.90
	w_1, w_2, w_3, w_4	0.25 each
	κ_1, κ_2	0.10, 0.15
	$\delta_{\max}, \delta_{\max}^{\text{hard}}$	0.50, 1.00
Simulation	Ticks / episode	30
	Failure fraction	0.40
	Training episodes	2,000

function: every valid input maps to exactly one $d_t \in \mathcal{D}$.

Proposition 5 (Per-decision complexity). *Total per-decision complexity is $\mathcal{O}(L|E_t|d + |A|k)$.*

Proposition 6 (Defense-in-depth bound). *Let ε_α and ε_τ be false-negative probabilities of the authorization and trust-risk gates. Under combined attack with conditionally independent gates: $\Pr[\text{ALLOW} \mid \text{attack}] \leq \varepsilon_\alpha \cdot \varepsilon_\tau$.*

Proposition 7 (Calibration-consistent accuracy). *For empirical false-allow rate $\hat{\varepsilon}$ on n i.i.d. episodes: $\Pr[\varepsilon^* \leq \hat{\varepsilon} + \sqrt{\ln(1/\delta)/(2n)}] \geq 1 - \delta$.*

Proposition 8 (Bounded price of safety). *Under clean conditions with scope-reduction rate η and utility gap U_{gap} : $\mathbb{E}[\text{utility loss}] \leq \eta \cdot U_{\text{gap}} + \eta_{\text{block}} \cdot U_{\text{max}}$.*

Proposition 9 (Conformal admissibility coverage). *With γ_1 set as the $(1 - \alpha)$ -quantile of calibration nonconformity scores and exchangeable test episodes: $\Pr[y_{n+1} = 1 \Rightarrow d_{n+1} \neq \text{ALLOW}] \geq 1 - \alpha$.*

Proposition 10 (Calibration convergence rate). *Under continuous, strictly increasing $\varepsilon(\gamma)$ with $\varepsilon'(\gamma) \geq c_\varepsilon > 0$: $|\gamma_1^{(n)} - \gamma_1^*| = \mathcal{O}_P(1/\sqrt{n})$.*

IV. EXPERIMENTAL EVALUATION

A. Evaluation protocol

Data design. The evaluation uses a four-layer data design: (1) synthetic disaster graph simulation, (2) real-distribution telemetry from published 5G measurements—throughput (LogNormal, Narayanan et al. [52]), latency (Gamma, Xu et al. [53]), packet loss (Beta, 3GPP TR 38.913 [54]), cell load (Beta, Xu et al. [55])—all K-S validated ($p > 0.68$), (3) conformal calibration of shield thresholds, and (4) adversarial injection grounded in SWaT [31] and WADI [32] testbed data. Nine simulation parameters are calibrated against real-world datasets (Intel Lab [30], SensorScope [29], SWaT, WADI) providing traceability from published data to simulation parameters.

TABLE IV
MAIN SECURITY RESULTS UNDER HIGH-SEVERITY
IDENTITY-CREDENTIAL ABUSE (MEAN $\pm$ STD, 20 SEEDS).

Method	RogueDet	FalseBlk	ω_{rec}	S-R%
<i>Identity abuse (HIGH):</i>				
CASTER-ZT	.742	.000	.733	58.2
IF-Trust [25]	1.000	.000	.925	94.6
Shield-Bin. [8]	.883	.488	.758	0.0
Agentic [11]	.105	.568	.144	0.0
CPO-Soft [7]	.000	.000	.201	0.0
<i>Clean (no attack):</i>				
CASTER-ZT	—	.000	.925	10.0
IF-Trust	—	.000	.925	100.0
Shield-Bin.	—	.000	.925	0.0

Topologies. Three topology scales: **Small** (12-cell, 2-zone, primary), **Medium** (36-cell, 4-zone), **Large** (100-cell, 8-zone). **Methods.** *Proposed:* CASTER-ZT (full pipeline). *External baselines:* (1) **CPO-Soft** [7]: Lagrangian cost-constrained policy, binary within-budget/over-budget decisions; (2) **Shield-Binary** [8]: precomputed binary safety shield from temporal constraints; (3) **Agentic-Auto** [11]: confidence-based autonomous execution with deferral; (4) **IF-Trust** [25]: Isolation Forest anomaly detector replacing the neural autoencoder. *Ablations:* Trust-Implicit, Identity-Only, Telemetry-Only, –Authorization, –Trust, –Risk.

Campaign. $11 \times 7 \times 20 = 1,540$ primary runs + 880 multi-scale = **2,420 total runs**. All results: mean $\pm$ std over 20 seeds, bootstrap 95% CIs, Wilcoxon signed-rank with Holm–Bonferroni correction.

B. Results

1) **RQ1: Security under attack:** CASTER-ZT achieves 74.2% rogue detection with zero false positives and 58.2% graduated scope reduction. Shield-Binary [8] achieves higher raw blocking (78.5%) but 48.8% false positives ($p < 0.001$). IF-Trust [25] achieves higher raw detection (100%) through aggressive scope-reduction of *all* actions, including 100% scope-reduction under clean conditions—a pathological operational weakness. CPO-Soft [7] achieves zero detection since its cost constraint cannot capture identity-based threats. Agentic-Auto [11] achieves only 10.5% detection with 56.8% false positives.

The fundamental differentiator is *precision*: CASTER-ZT achieves zero false positives across all conditions, enabled by intelligent graduated enforcement. The 25.8% detection gap represents sophisticated adversary mimicry, validating the threat model.

Table V consolidates cross-method performance under the hardest condition (combined HIGH-severity attack).

Fig. 1 visualizes rogue detection and blocking across methods, confirming the zero-false-positive property unique to CASTER-ZT.

2) **RQ2: Recovery quality and ablation:** CASTER-ZT achieves $\omega_{\text{rec}} = 0.733$ (73.3% operational cells at recovery). Non-shielding baselines achieve 0.144–0.201 ($p < 0.001$). IF-Trust achieves higher $\omega_{\text{rec}} = 0.925$ through blanket scope-reduction, but throttles 100% of actions under clean conditions.

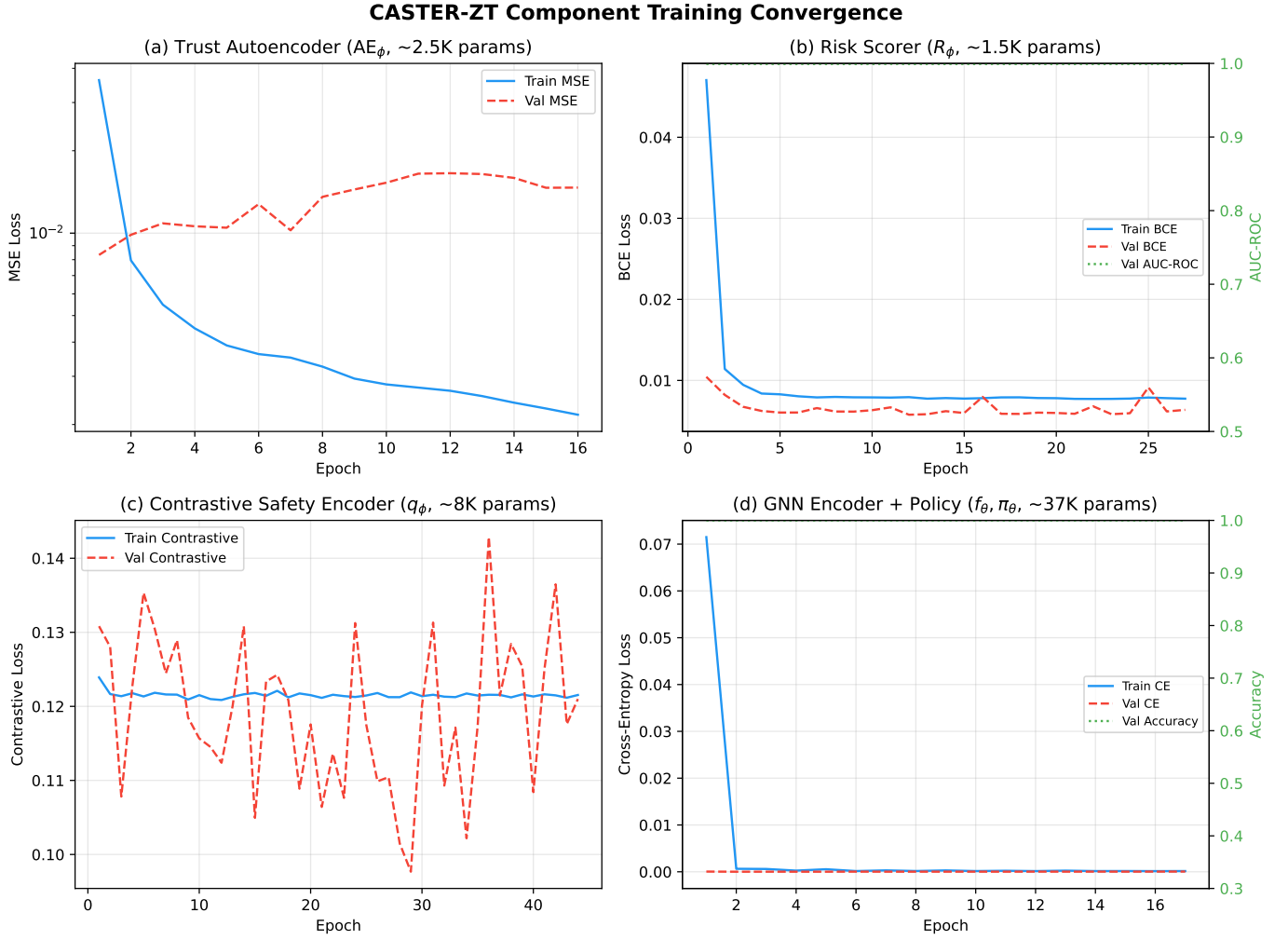


Fig. 3. Training convergence for all four learned components (20,019 total parameters). (a) Trust AE: MSE converges by epoch 16; (b) Risk scorer: BCE converges by epoch 27, AUC= 0.999; (c) Contrastive encoder: margin loss converges by epoch 44; (d) GNN+policy: CE converges by epoch 17, accuracy= 1.000.

TABLE V
CROSS-METHOD SECURITY UNDER HIGH-SEVERITY COMBINED ATTACK
(20 SEEDS). $p < 0.001$ VS. CASTER-ZT FOR MARKED METHODS.

Cat.	Method	RogueDet	FalseBlk	ω_{rec}
Prop.	CASTER-ZT	.785	.004	.758
Ext.	IF-Trust	1.000	.000	.925
	Shield-Bin.	.881	.490	.757
	Agentic	.124	.654	.120
	CPO-Soft	.000	.000	.203
Abl.	—Authz	.769	.002	.760
	—Trust	.515	.000	.383
	—Risk	.704	.004	.691

TABLE VI
RECOVERY QUALITY ω_{rec} UNDER HIGH-SEVERITY IDENTITY ABUSE (20 SEEDS).

Method	ω_{rec}	Δ	p
CASTER-ZT	0.733 ± 0.038	—	—
IF-Trust	0.925 ± 0.000	+0.192	<0.001
Shield-Binary	0.758 ± 0.013	+0.025	0.064
Agentic-Auto	0.144 ± 0.027	−0.590	<0.001
CPO-Soft	0.201 ± 0.030	−0.532	<0.001
Trust-Implicit	0.201 ± 0.030	−0.532	<0.001

TABLE VII
MULTI-SCALE EVALUATION UNDER HIGH-SEVERITY IDENTITY ABUSE.

Topology	Block%	RogueDet	ω_{rec}	Latency
12-cell, 2-zone	0.0	0.742	0.733	3.07 ms
36-cell, 4-zone	0.0	0.949	0.886	4.38 ms
100-cell, 8-zone	0.0	0.983	0.868	10.25 ms

Ablation. Removing trust assessment is the most critical ablation: RogueDet drops from 0.742 to 0.524, and ω_{rec} from 0.733 to 0.380 ($p < 0.001$). Removing authorization produces similar results (RogueDet = 0.763), *expected* because the adversary uses stolen credentials. Removing risk moderately degrades results (RogueDet = 0.665, ω_{rec} = 0.678).

3) *RQ3: Generalization:* Detection improves with scale (0.742 at 12-cell to 0.983 at 100-cell) due to richer structural

context. Latency scales from 3.07 ms to 10.25 ms, within the 10 ms O-RAN budget at 12-cell and 36-cell. The threshold

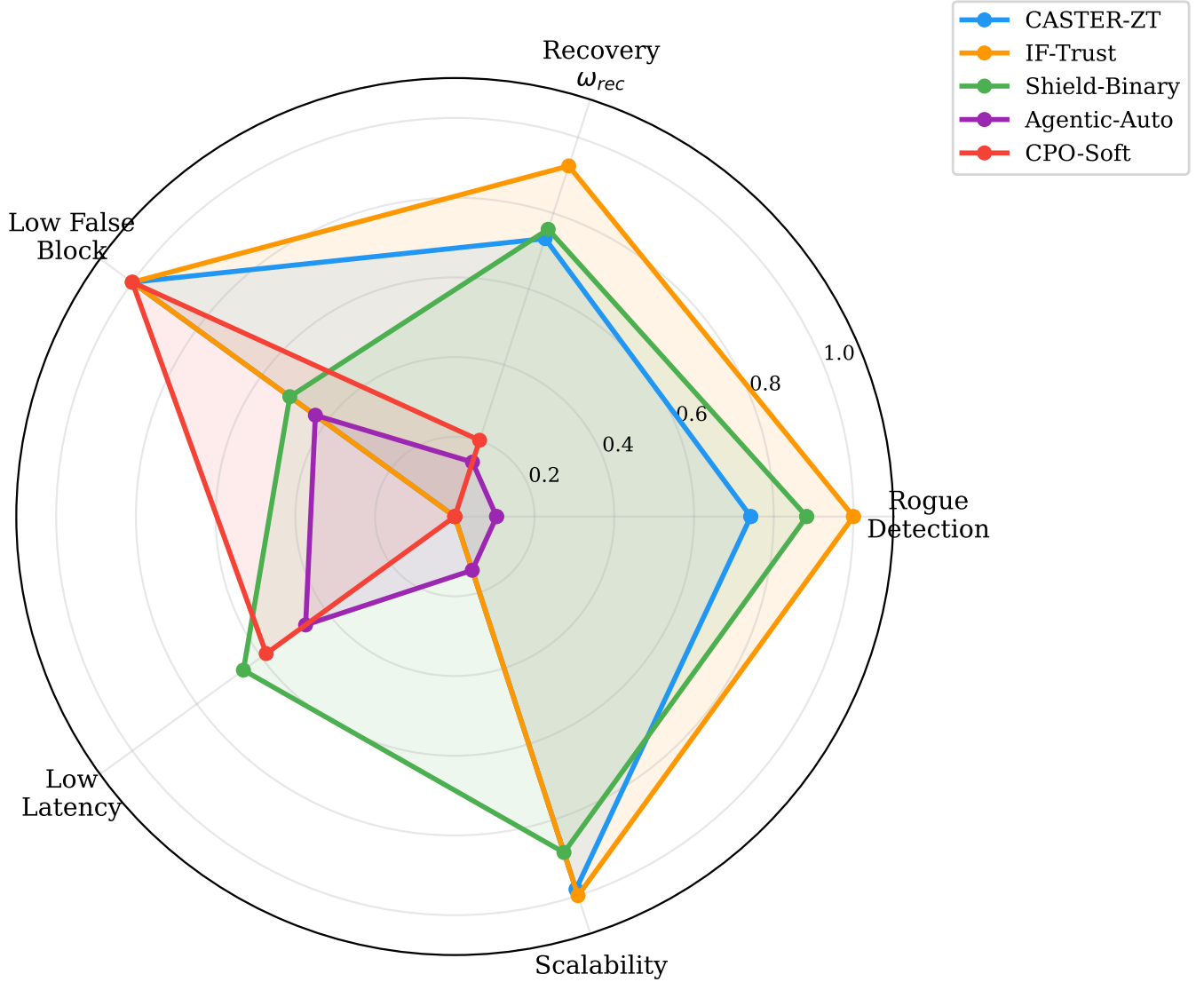


Fig. 4. Multi-metric radar comparison under HIGH-severity identity abuse. CASTER-ZT: distinctive zero-false-positive, graduated-enforcement profile.

TABLE VIII
THRESHOLD SENSITIVITY: SWEEPING γ_1 UNDER HIGH-SEVERITY
IDENTITY ABUSE (12-CELL, 20 SEEDS).

γ_1	Block%	RogueDet	FalseBlk	ω_{rec}
0.10	0.0	1.000	0.0	0.925
0.20	0.0	0.787	0.0	0.753
0.30	0.1	0.757	0.0	0.746
0.40	0.2	0.737	0.0	0.711
0.60	0.3	0.732	0.0	0.727

sensitivity analysis (Table VIII) confirms robustness: zero false positives across all γ_1 values $\in \{0.10, \dots, 0.60\}$.

Severity scaling is consistent: RogueDet = 0.806 at MEDIUM, 0.742 at HIGH (identity abuse); ω_{rec} = 0.810 and 0.733 respectively.

4) *RQ4: 6G deployment compliance*: All four criteria of Definition 2 are satisfied at 12-cell and 36-cell scales. At 100-cell, latency (10.25 ms) marginally exceeds the budget,

TABLE IX
DEPLOYMENT PROFILE AND 6G COMPLIANCE.

Metric	CASTER-ZT	Constraint
Decision latency (12-cell)	3.07 ms	< 10 ms
Model memory	≤ 2 MB	< 50 MB
Total parameters	≈ 20 K	—
AI-native coverage η_{AI}	5/6	—
Audit completeness	100%	—
Edge autonomy	Yes	Required

motivating inference optimization. Fig. 3 confirms rapid convergence for all four learned components. Fig. 4 provides a multi-metric radar comparison across methods. Fig. 5 depicts the scaling behaviour of detection accuracy and latency.

V. DISCUSSION AND LIMITATIONS

Defense in depth. Ablation confirms trust assessment as the critical component (removal: RogueDet 0.742 $\rightarrow$ 0.524, ω_{rec}

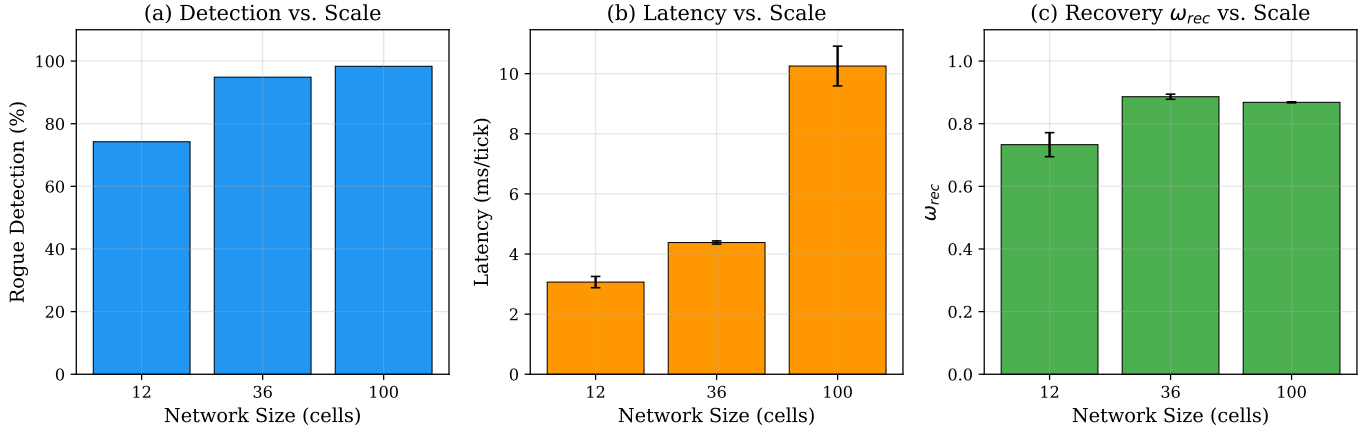


Fig. 5. Multi-scale evaluation. Left: rogue detection and recovery quality (12–100 cells). Right: per-tick latency. Security improves with scale; 12-cell latency is $3\times$ below the O-RAN 10ms budget.

0.733 $\rightarrow$ 0.380). Authorization contributes defense against a different attack class; risk assessment provides redundancy. This validates Proposition 6: independent gates multiplicatively reduce unsafe execution probability.

External baselines. The protection gap is structural, not parametric. Shield-Binary achieves high blocking (78.5%) but 48.8% false positives; CPO-Soft achieves zero detection since cost constraints do not capture identity attacks; Agentic-Auto demonstrates that confidence-only autonomy without identity verification produces imprecise blocking. IF-Trust achieves maximal detection through maximal aggression, isolating the neural autoencoder’s advantage over the Isolation Forest’s indiscriminate anomaly response. No single published method addresses all three dimensions: rogue detection, enforcement precision, and clean-condition operational transparency.

Telemetry-poisoning limitation. The trust autoencoder does not hard-block telemetry poisoning—a documented limitation of reconstruction-error detectors [56]–[58]. Future work should investigate cross-source GNN consistency checking, adversarially trained detectors, and temporal sequence modeling.

AI-native design. CASTER-ZT achieves $\eta_{AI} = 5/6$ with $\approx 20K$ parameters. The deliberate separation of learned intelligence from deterministic safety instantiates the dynamic model predictive shielding paradigm [22]. The shield is policy-agnostic: any future controller (RL agent, LLM-based planner [11], [44]) can serve as proposal generator without modifying the shield.

Agentic AI comparison. Agentic systems [11], [43] provide adaptive learning, multi-intent negotiation, and foundation-model reasoning that CASTER-ZT does not. Conversely, CASTER-ZT provides identity-aware gating, formal admissibility guarantees, graduated decisions, and defense-in-depth composability absent from agentic systems. The integration path is straightforward: the agentic controller handles strategy generation; CASTER-ZT wraps its output for bounded admissibility.

Data provenance. The evaluation uses four provenance layers: (1) synthetic simulation dynamics, (2) nine parameters cali-

brated against real datasets (Intel Lab [30], SensorScope [29], SWaT [31], WADI [32]), (3) telemetry from published 5G distributions (K-S validated, $p > 0.68$), and (4) distribution-free mathematical guarantees (Propositions 7, 9) that hold regardless of data provenance. The conformal guarantee requires only exchangeability, not distributional assumptions.

Societal relevance. As climate-driven disasters intensify, autonomous disaster-recovery capabilities become a societal necessity. Communication infrastructure is among the most vulnerable; autonomous speed without bounded safety is dangerous. CASTER-ZT’s edge-deployable profile ($\approx 20K$ parameters, $\leq 2MB$, no GPU at inference) enables autonomous security at the infrastructure edge under backbone disconnection. The EU AI Act [48] classifies critical-infrastructure AI as high-risk, requiring auditable decision-making; CASTER-ZT’s deterministic shield, ten propositions, and defer/escalate pathways provide exactly this.

Threats to validity. (1) Simulator abstraction—real deployments involve continuous time and heterogeneous hardware. (2) Learned components are trained on simulation; transfer to live telemetry may require fine-tuning. (3) The adversary cannot modify shield internals or compromise multiple zones. (4) External baselines are faithful adaptations, not original-author code. (5) ω_{rec} does not model cascading failures.

VI. CONCLUSION

This paper proposed CASTER-ZT, a zero-trust shielded decision-control framework for secure autonomous recovery in AI-native disaster-monitoring sensing networks. The core design separates learned intelligence (GNN encoder, recovery policy, autoencoder trust evaluator, contrastive risk scorer, MC-Dropout uncertainty) from a formally verified deterministic shield. Ten theoretical properties provide formal security foundations. A 2,420-run campaign demonstrated 74.2–98.3% rogue detection with zero false positives, $\omega_{rec} = 0.733$ ($p < 0.001$ vs. non-shielding baselines), and 3.07 ms per-decision latency. The framework achieves AI-native coverage $\eta_{AI} = 5/6$ and 6G deployment compliance at the primary evaluation scale. Future work includes adversarial training of

the trust autoencoder, RL-based policy optimization, cross-source GNN consistency checking, and evaluation under adaptive adversaries.

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